

AJKD Atlas of Renal Pathology: Collapsing Glomerulopathy



Agnes B. Fogo, MD,¹ Mark A. Lusco, MD,¹ Behzad Najafian, MD,² and Charles E. Alpers, MD²

Clinical and Pathologic Features

Collapsing glomerulopathy is associated with marked proteinuria, rapid decrease in kidney function, and a poor prognosis. This lesion is unresponsive to corticosteroid therapy alone. There is a strong preponderance in African Americans versus whites with collapsing glomerulopathy.

Light microscopy: Segmental or global glomerular tuft collapse with overlying visceral epithelial hyperplasia and hypertrophy. Hypertrophied visceral epithelial cells have frequent protein droplets.

Immunofluorescence microscopy: No or limited deposits (nonspecific immunoglobulin M and C3 staining in collapsed segments).

Electron microscopy: Glomerular basement membranes are wrinkled and collapsed. Overlying visceral epithelial cells show hypertrophy and hyperplasia, with frequent vacuoles and protein droplets. There is extensive foot process effacement and no or limited mesangial deposits. HIV (human immunodeficiency virus)-associated nephropathy or lupus nephritis-associated collapsing lesions show reticular aggregates, in contrast to idiopathic collapsing glomerulopathy, no reticular aggregates are present.

Etiology/Pathogenesis

The etiology of collapsing glomerulopathy has not yet been discovered. There is proliferation and dedifferentiation of mature podocytes with loss of cyclin-dependent kinase inhibitor 1B (p27Kip1) expression in areas of collapse. Viral agents (HIV and parvovirus) have been proposed as a possible etiology. Treatment with pamidronate or calcineurin inhibitors has been linked to collapsing glomerulopathy. Patients with a risk allele variant of the apolipoprotein APOL1,

common in African Americans, have an increased incidence of collapsing glomerulopathy.

Differential Diagnosis

Crescentic glomerulonephritis has parietal epithelial proliferation, but generally does not have reabsorption droplets. Crescents also often show fibrinoid necrosis and glomerular basement membrane breaks, features that distinguish them from collapsing lesions.

Secondary collapsing glomerulopathy can be related to infection (eg, HIV and parvovirus), drugs (eg, bisphosphonate and calcineurin inhibitors), severe vascular disease (eg, thrombotic microangiopathy and cocaine use), and autoimmune disease (eg, systemic lupus erythematosus).

Key Diagnostic Features

- Extensive foot process effacement
- Absence of immune deposits
- Diagnostic collapsing lesion, with collapse of tuft and visceral epithelial cell hyperplasia

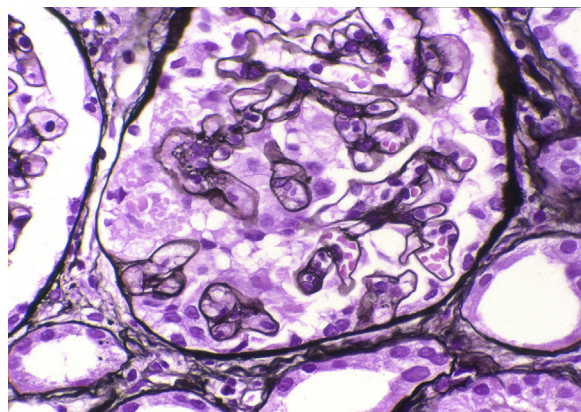


Figure 1. Collapsing glomerulopathy is characterized by collapse of the glomerular tuft with marked proliferation of overlying glomerular visceral epithelial cells, often with prominent protein droplets, as shown here (Jones silver stain). Reproduced with permission from AJKD 38(5):e24.

From the ¹Department of Pathology, Microbiology and Immunology, Vanderbilt University, Nashville, TN; and ²Department of Pathology, University of Washington, Seattle, WA.

Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

Address correspondence to agnes.fogo@vanderbilt.edu

Am J Kidney Dis. 66(2):e3-e4.

© 2015 by the National Kidney Foundation, Inc.

0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2015.04.009>

Continued

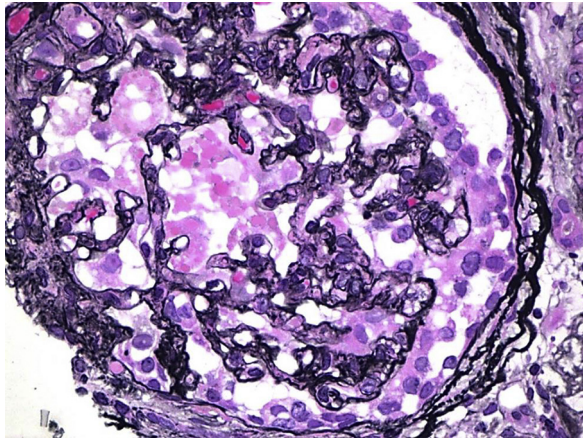


Figure 2. The collapsing pattern in collapsing glomerulopathy is evident here, with accentuation of each lobule due to the collapse of the tuft. There is also overlying glomerular visceral epithelial cell hyperplasia with protein droplets. Occasional intracapillary foam cells are also evident (Jones silver stain).

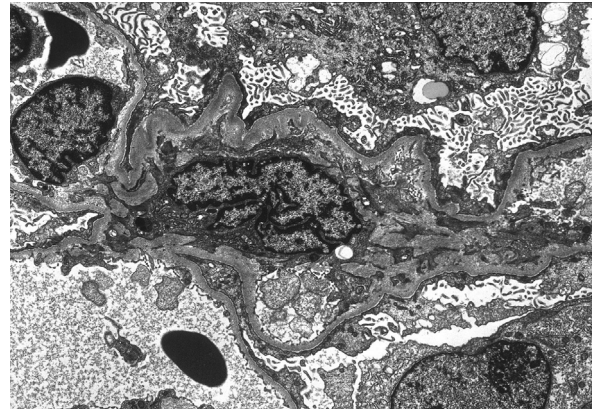


Figure 3. In idiopathic collapsing glomerulopathy, the collapse of the tuft is evident by electron microscopy. There is extensive foot process effacement with microvillous change. This is associated with loss of podocyte differentiation markers of the glomerular visceral epithelial cells. Reproduced with permission from *AJKD* 38(5):e24.